

ANALYTICAL REPORT

Eurofins TestAmerica, Michigan
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Laboratory Job ID: 190-27106-1

Client Project/Site: City of Cadillac- Semiannual PFAS

For:

City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:
10/26/2021 9:55:18 AM

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Results relate only to the items tested and the sample(s) as received by the laboratory.

Table of Contents

Cover Page	1
Table of Contents	2
Sample Summary	3
Case Narrative	4
Client Sample Results	5
QC Sample Results	11
Isotope Dilution Summary	15
Definitions/Glossary	16
QC Association Summary	17
Lab Chronicle	18
Certification Summary	19
Chain of Custody	20



Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-27106-1	Field Blank	Water	10/13/21 14:10	10/14/21 09:46
190-27106-2	Outfall	Water	10/13/21 14:15	10/14/21 09:46
190-27106-3	Outfall Duplicate	Water	10/13/21 14:17	10/14/21 09:46

Case Narrative

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Job ID: 190-27106-1

Laboratory: Eurofins TestAmerica, Michigan

Narrative

Job Narrative 190-27106-1

Comments

The PFC analysis was performed at the Eurofins TestAmerica Sacramento laboratory.

Receipt

The samples were received on 10/14/2021 9:46 AM. Unless otherwise noted below, the samples arrived in good condition, and where required, properly preserved and on ice. The temperature of the cooler at receipt was 4.3° C.

LCMS

Method 537 (modified): Isotope Dilution Analyte (IDA) recovery is above the method recommended limit for the following samples: Outfall (190-27106-2) and Outfall Duplicate (190-27106-3). Quantitation by isotope dilution generally precludes any adverse effect on data quality due to elevated IDA recoveries.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Organic Prep

Method 3535: Insufficient sample volume was available to perform a matrix spike/matrix spike duplicate (MS/MSD) associated with preparation batch 320-535560.

320-535560

Method: 3535 PFC-W

Method 3535: The following samples contained floating particulates in the sample bottle prior to extraction: Outfall (190-27106-2) and Outfall Duplicate (190-27106-3).

320-535560

Method: 3535 PFC-W

Method 3535: The following samples were light yellow after final volume: Outfall (190-27106-2) and Outfall Duplicate (190-27106-3).

320-535560

Method: 3535 PFC-W

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Client Sample ID: Field Blank

Lab Sample ID: 190-27106-1

Date Collected: 10/13/21 14:10

Matrix: Water

Date Received: 10/14/21 09:46

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
F-53B Major	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
F-53B Minor	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
4:2 FTS	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
6:2 FTS	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:04	1
8:2 FTS	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
HFPO-DA (GenX)	<3.8		3.8	ng/L		10/20/21 05:37	10/22/21 14:04	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:04	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorobutanesulfonic acid (PFBS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorobutanoic acid (PFBA)	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorodecanesulfonic acid (PFDS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorodecanoic acid (PFDA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorododecanoic acid (PFDoA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluoroheptanoic acid (PFHpA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorohexanesulfonic acid (PFHxS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorohexanoic acid (PFHxA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorononanesulfonic acid (PFNS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorononanoic acid (PFNA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorooctanesulfonamide (FOSA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorooctanesulfonic acid (PFOS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorooctanoic acid (PFOA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluoropentanesulfonic acid (PFPeS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluoropentanoic acid (PFPeA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorotetradecanoic acid (PFTeA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluorotridecanoic acid (PFTriA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1
Perfluoroundecanoic acid (PFUnA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:04	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	76		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C3 HFPO-DA	82		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C4 PFBA	89		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C3 PFBS	79		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C2 PFDA	87		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C2 PFDoA	83		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C4 PFHpA	89		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C2 PFHxA	90		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C5 PFNA	90		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C4 PFOA	92		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C4 PFOS	93		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C5 PFPeA	94		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C2 PFTeDA	85		25 - 150	10/20/21 05:37	10/22/21 14:04	1
13C2 PFUnA	80		25 - 150	10/20/21 05:37	10/22/21 14:04	1
d5-NEtFOSAA	88		25 - 150	10/20/21 05:37	10/22/21 14:04	1
d3-NMeFOSAA	86		25 - 150	10/20/21 05:37	10/22/21 14:04	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Client Sample ID: Field Blank

Date Collected: 10/13/21 14:10

Date Received: 10/14/21 09:46

Lab Sample ID: 190-27106-1

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
M2-4:2 FTS	83		25 - 150	10/20/21 05:37	10/22/21 14:04	1
M2-6:2 FTS	94		25 - 150	10/20/21 05:37	10/22/21 14:04	1
M2-8:2 FTS	89		25 - 150	10/20/21 05:37	10/22/21 14:04	1
18O2 PFHxS	89		25 - 150	10/20/21 05:37	10/22/21 14:04	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Client Sample ID: Outfall

Lab Sample ID: 190-27106-2

Date Collected: 10/13/21 14:15

Matrix: Water

Date Received: 10/14/21 09:46

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
F-53B Major	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
F-53B Minor	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
4:2 FTS	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
6:2 FTS	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:35	1
8:2 FTS	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
HFPO-DA (GenX)	<3.8		3.8	ng/L		10/20/21 05:37	10/22/21 14:35	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:35	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorobutanesulfonic acid (PFBS)	19		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorobutanoic acid (PFBA)	<4.7		4.7	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorodecanesulfonic acid (PFDS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorodecanoic acid (PFDA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorododecanoic acid (PFDoA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluoroheptanoic acid (PFHpA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorohexanesulfonic acid (PFHxS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorohexanoic acid (PFHxA)	12		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorononanesulfonic acid (PFNS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorononanoic acid (PFNA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorooctanesulfonamide (FOSA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorooctanesulfonic acid (PFOS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorooctanoic acid (PFOA)	2.0		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluoropentanesulfonic acid (PFPeS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluoropentanoic acid (PFPeA)	3.3		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorotetradecanoic acid (PFTeA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluorotridecanoic acid (PFTriA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1
Perfluoroundecanoic acid (PFUnA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:35	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	81		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C3 HFPO-DA	76		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C4 PFBA	78		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C3 PFBS	86		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C2 PFDA	89		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C2 PFDoA	77		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C4 PFHpA	90		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C2 PFHxA	70		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C5 PFNA	96		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C4 PFOA	94		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C4 PFOS	89		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C5 PFPeA	88		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C2 PFTeDA	56		25 - 150	10/20/21 05:37	10/22/21 14:35	1
13C2 PFUnA	84		25 - 150	10/20/21 05:37	10/22/21 14:35	1
d5-NEtFOSAA	103		25 - 150	10/20/21 05:37	10/22/21 14:35	1
d3-NMeFOSAA	90		25 - 150	10/20/21 05:37	10/22/21 14:35	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Client Sample ID: Outfall

Date Collected: 10/13/21 14:15

Date Received: 10/14/21 09:46

Lab Sample ID: 190-27106-2

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
M2-4:2 FTS	125		25 - 150	10/20/21 05:37	10/22/21 14:35	1
M2-6:2 FTS	163	*5+	25 - 150	10/20/21 05:37	10/22/21 14:35	1
M2-8:2 FTS	155	*5+	25 - 150	10/20/21 05:37	10/22/21 14:35	1
18O2 PFHxS	87		25 - 150	10/20/21 05:37	10/22/21 14:35	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Client Sample ID: Outfall Duplicate

Lab Sample ID: 190-27106-3

Date Collected: 10/13/21 14:17

Matrix: Water

Date Received: 10/14/21 09:46

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
F-53B Major	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
F-53B Minor	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
4:2 FTS	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
6:2 FTS	<4.8		4.8	ng/L		10/20/21 05:37	10/22/21 14:46	1
8:2 FTS	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
HFPO-DA (GenX)	<3.9		3.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<4.8		4.8	ng/L		10/20/21 05:37	10/22/21 14:46	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<4.8		4.8	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorobutanesulfonic acid (PFBS)	20		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorobutanoic acid (PFBA)	<4.8		4.8	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorodecanesulfonic acid (PFDS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorodecanoic acid (PFDA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorododecanoic acid (PFDoA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluoroheptanoic acid (PFHpA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorohexanesulfonic acid (PFHxS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorohexanoic acid (PFHxA)	12		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorononanesulfonic acid (PFNS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorononanoic acid (PFNA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorooctanesulfonamide (FOSA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorooctanesulfonic acid (PFOS)	2.0		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorooctanoic acid (PFOA)	2.3		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluoropentanesulfonic acid (PFPeS)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluoropentanoic acid (PFPeA)	3.8		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorotetradecanoic acid (PFTeA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluorotridecanoic acid (PFTriA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1
Perfluoroundecanoic acid (PFUnA)	<1.9		1.9	ng/L		10/20/21 05:37	10/22/21 14:46	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	83		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C3 HFPO-DA	85		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C4 PFBA	77		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C3 PFBS	83		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C2 PFDA	87		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C2 PFDoA	80		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C4 PFHpA	92		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C2 PFHxA	71		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C5 PFNA	93		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C4 PFOA	96		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C4 PFOS	87		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C5 PFPeA	86		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C2 PFTeDA	55		25 - 150	10/20/21 05:37	10/22/21 14:46	1
13C2 PFUnA	88		25 - 150	10/20/21 05:37	10/22/21 14:46	1
d5-NEtFOSAA	106		25 - 150	10/20/21 05:37	10/22/21 14:46	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Client Sample ID: Outfall Duplicate

Lab Sample ID: 190-27106-3

Date Collected: 10/13/21 14:17

Matrix: Water

Date Received: 10/14/21 09:46

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
d3-NMeFOSAA	92		25 - 150	10/20/21 05:37	10/22/21 14:46	1
M2-4:2 FTS	133		25 - 150	10/20/21 05:37	10/22/21 14:46	1
M2-6:2 FTS	150		25 - 150	10/20/21 05:37	10/22/21 14:46	1
M2-8:2 FTS	151	*5+	25 - 150	10/20/21 05:37	10/22/21 14:46	1
18O2 PFHxS	87		25 - 150	10/20/21 05:37	10/22/21 14:46	1

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 320-535560/1-A

Matrix: Water

Analysis Batch: 536189

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 535560

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
F-53B Major	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
F-53B Minor	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
4:2 FTS	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
6:2 FTS	<5.0		5.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
8:2 FTS	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
HFPO-DA (GenX)	<4.0		4.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<5.0		5.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<5.0		5.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorobutanoic acid (PFBA)	<5.0		5.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorooctanesulfonic acid (PFOS)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		10/20/21 05:37	10/22/21 12:30	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	79		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C3 HFPO-DA	88		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C4 PFBA	90		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C3 PFBS	87		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C2 PFDA	86		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C2 PFDoA	85		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C4 PFHpA	92		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C2 PFHxA	92		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C5 PFNA	95		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C4 PFOA	89		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C4 PFOS	88		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C5 PFPeA	90		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C2 PFTeA	87		25 - 150	10/20/21 05:37	10/22/21 12:30	1
13C2 PFUnA	88		25 - 150	10/20/21 05:37	10/22/21 12:30	1
d5-NEtFOSAA	96		25 - 150	10/20/21 05:37	10/22/21 12:30	1

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 320-535560/1-A

Matrix: Water

Analysis Batch: 536189

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 535560

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
d3-NMeFOSAA	93		25 - 150	10/20/21 05:37	10/22/21 12:30	1
M2-4:2 FTS	92		25 - 150	10/20/21 05:37	10/22/21 12:30	1
M2-6:2 FTS	92		25 - 150	10/20/21 05:37	10/22/21 12:30	1
M2-8:2 FTS	93		25 - 150	10/20/21 05:37	10/22/21 12:30	1
18O2 PFHxS	88		25 - 150	10/20/21 05:37	10/22/21 12:30	1

Lab Sample ID: LCS 320-535560/2-A

Matrix: Water

Analysis Batch: 536189

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 535560

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	37.7	34.5		ng/L		92	79 - 139
F-53B Major	37.3	36.6		ng/L		98	75 - 135
F-53B Minor	37.7	35.5		ng/L		94	54 - 114
4:2 FTS	37.4	37.7		ng/L		101	79 - 139
6:2 FTS	37.9	36.5		ng/L		96	59 - 175
8:2 FTS	38.3	39.2		ng/L		102	75 - 135
HFPO-DA (GenX)	40.0	36.6		ng/L		92	51 - 173
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	45.0		ng/L		112	76 - 136
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	42.1		ng/L		105	76 - 136
Perfluorobutanesulfonic acid (PFBS)	35.4	35.4		ng/L		100	67 - 127
Perfluorobutanoic acid (PFBA)	40.0	40.9		ng/L		102	76 - 136
Perfluorodecanesulfonic acid (PFDS)	38.6	37.6		ng/L		98	71 - 131
Perfluorodecanoic acid (PFDA)	40.0	37.7		ng/L		94	76 - 136
Perfluorododecanoic acid (PFDoA)	40.0	42.1		ng/L		105	71 - 131
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	37.0		ng/L		97	76 - 136
Perfluoroheptanoic acid (PFHpA)	40.0	42.0		ng/L		105	72 - 132
Perfluorohexanesulfonic acid (PFHxS)	36.4	37.6		ng/L		103	59 - 119
Perfluorohexanoic acid (PFHxA)	40.0	42.3		ng/L		106	73 - 133
Perfluorononanesulfonic acid (PFNS)	38.4	41.3		ng/L		108	75 - 135
Perfluorononanoic acid (PFNA)	40.0	43.1		ng/L		108	75 - 135
Perfluorooctanesulfonamide (FOSA)	40.0	41.6		ng/L		104	73 - 133
Perfluorooctanesulfonic acid (PFOS)	37.1	35.0		ng/L		94	70 - 130
Perfluorooctanoic acid (PFOA)	40.0	41.2		ng/L		103	70 - 130
Perfluoropentanesulfonic acid (PFPeS)	37.5	40.3		ng/L		108	66 - 126
Perfluoropentanoic acid (PFPeA)	40.0	38.4		ng/L		96	71 - 131
Perfluorotetradecanoic acid (PFTeA)	40.0	40.0		ng/L		100	70 - 130
Perfluorotridecanoic acid (PFTriA)	40.0	43.1		ng/L		108	71 - 131

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-535560/2-A

Matrix: Water

Analysis Batch: 536189

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 535560

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluoroundecanoic acid (PFUnA)	40.0	42.4		ng/L		106	68 - 128
Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits				
13C8 FOSA	69		25 - 150				
13C3 HFPO-DA	85		25 - 150				
13C4 PFBA	78		25 - 150				
13C3 PFBS	81		25 - 150				
13C2 PFDA	76		25 - 150				
13C2 PFDoA	77		25 - 150				
13C4 PFHpA	80		25 - 150				
13C2 PFHxA	78		25 - 150				
13C5 PFNA	79		25 - 150				
13C4 PFOA	78		25 - 150				
13C4 PFOS	83		25 - 150				
13C5 PFPeA	81		25 - 150				
13C2 PFTeDA	78		25 - 150				
13C2 PFUnA	75		25 - 150				
d5-NEtFOSAA	80		25 - 150				
d3-NMeFOSAA	80		25 - 150				
M2-4:2 FTS	89		25 - 150				
M2-6:2 FTS	83		25 - 150				
M2-8:2 FTS	77		25 - 150				
18O2 PFHxS	79		25 - 150				

Lab Sample ID: LCSD 320-535560/3-A

Matrix: Water

Analysis Batch: 536189

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 535560

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
4,8-Dioxa-3H-perfluorononanoic acid (ADONA)	37.7	38.6		ng/L		103	79 - 139	11	30
F-53B Major	37.3	38.9		ng/L		104	75 - 135	6	30
F-53B Minor	37.7	36.5		ng/L		97	54 - 114	3	30
4:2 FTS	37.4	36.5		ng/L		98	79 - 139	3	30
6:2 FTS	37.9	36.0		ng/L		95	59 - 175	1	30
8:2 FTS	38.3	38.0		ng/L		99	75 - 135	3	30
HFPO-DA (GenX)	40.0	38.9		ng/L		97	51 - 173	6	30
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	40.7		ng/L		102	76 - 136	10	30
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	40.5		ng/L		101	76 - 136	4	30
Perfluorobutanesulfonic acid (PFBS)	35.4	36.7		ng/L		104	67 - 127	3	30
Perfluorobutanoic acid (PFBA)	40.0	40.3		ng/L		101	76 - 136	1	30
Perfluorodecanesulfonic acid (PFDS)	38.6	39.7		ng/L		103	71 - 131	6	30
Perfluorodecanoic acid (PFDA)	40.0	34.1		ng/L		85	76 - 136	10	30
Perfluorododecanoic acid (PFDoA)	40.0	46.3		ng/L		116	71 - 131	10	30

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCSD 320-535560/3-A

Matrix: Water

Analysis Batch: 536189

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 535560

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	39.2		ng/L		103	76 - 136	6	30
Perfluoroheptanoic acid (PFHpA)	40.0	42.6		ng/L		107	72 - 132	2	30
Perfluorohexanesulfonic acid (PFHxS)	36.4	37.5		ng/L		103	59 - 119	0	30
Perfluorohexanoic acid (PFHxA)	40.0	38.9		ng/L		97	73 - 133	8	30
Perfluorononanesulfonic acid (PFNS)	38.4	41.3		ng/L		108	75 - 135	0	30
Perfluorononanoic acid (PFNA)	40.0	41.0		ng/L		103	75 - 135	5	30
Perfluorooctanesulfonamide (FOSA)	40.0	42.2		ng/L		105	73 - 133	1	30
Perfluorooctanesulfonic acid (PFOS)	37.1	36.8		ng/L		99	70 - 130	5	30
Perfluorooctanoic acid (PFOA)	40.0	39.6		ng/L		99	70 - 130	4	30
Perfluoropentanesulfonic acid (PFPeS)	37.5	40.3		ng/L		107	66 - 126	0	30
Perfluoropentanoic acid (PFPeA)	40.0	37.8		ng/L		94	71 - 131	2	30
Perfluorotetradecanoic acid (PFTeA)	40.0	38.6		ng/L		96	70 - 130	4	30
Perfluorotridecanoic acid (PFTriA)	40.0	44.7		ng/L		112	71 - 131	4	30
Perfluoroundecanoic acid (PFUnA)	40.0	43.2		ng/L		108	68 - 128	2	30

Isotope Dilution	LCSD %Recovery	LCSD Qualifier	LCSD Limits
13C8 FOSA	74		25 - 150
13C3 HFPO-DA	88		25 - 150
13C4 PFBA	83		25 - 150
13C3 PFBS	86		25 - 150
13C2 PFDA	83		25 - 150
13C2 PFDoA	80		25 - 150
13C4 PFHpA	81		25 - 150
13C2 PFHxA	85		25 - 150
13C5 PFNA	89		25 - 150
13C4 PFOA	87		25 - 150
13C4 PFOS	85		25 - 150
13C5 PFPeA	95		25 - 150
13C2 PFTeDA	85		25 - 150
13C2 PFUnA	84		25 - 150
d5-NEtFOSAA	93		25 - 150
d3-NMeFOSAA	90		25 - 150
M2-4:2 FTS	98		25 - 150
M2-6:2 FTS	91		25 - 150
M2-8:2 FTS	92		25 - 150
18O2 PFHxS	82		25 - 150

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (25-150)	PFDA (25-150)	PFDoA (25-150)	C4PFHA (25-150)	PFHxA (25-150)
190-27106-1	Field Blank	76	82	89	79	87	83	89	90
190-27106-2	Outfall	81	76	78	86	89	77	90	70
190-27106-3	Outfall Duplicate	83	85	77	83	87	80	92	71
LCS 320-535560/2-A	Lab Control Sample	69	85	78	81	76	77	80	78
LCSD 320-535560/3-A	Lab Control Sample Dup	74	88	83	86	83	80	81	85
MB 320-535560/1-A	Method Blank	79	88	90	87	86	85	92	92

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (25-150)	PFOA (25-150)	PFOS (25-150)	PFPeA (25-150)	PFTDA (25-150)	PFUnA (25-150)	d5NEFOS (25-150)	d3NMFOS (25-150)
190-27106-1	Field Blank	90	92	93	94	85	80	88	86
190-27106-2	Outfall	96	94	89	88	56	84	103	90
190-27106-3	Outfall Duplicate	93	96	87	86	55	88	106	92
LCS 320-535560/2-A	Lab Control Sample	79	78	83	81	78	75	80	80
LCSD 320-535560/3-A	Lab Control Sample Dup	89	87	85	95	85	84	93	90
MB 320-535560/1-A	Method Blank	95	89	88	90	87	88	96	93

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (25-150)
190-27106-1	Field Blank	83	94	89	89
190-27106-2	Outfall	125	163 *5+	155 *5+	87
190-27106-3	Outfall Duplicate	133	150	151 *5+	87
LCS 320-535560/2-A	Lab Control Sample	89	83	77	79
LCSD 320-535560/3-A	Lab Control Sample Dup	98	91	92	82
MB 320-535560/1-A	Method Blank	92	92	93	88

Surrogate Legend

PFOSA = 13C8 FOSA
HFPODA = 13C3 HFPO-DA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5 PFPeA
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*5+	Isotope dilution analyte is outside acceptance limits, high biased.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
α	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CFU	Colony Forming Unit
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MCL	EPA recommended "Maximum Contaminant Level"
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
MPN	Most Probable Number
MQL	Method Quantitation Limit
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
NEG	Negative / Absent
POS	Positive / Present
PQL	Practical Quantitation Limit
PRES	Presumptive
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)
TNTC	Too Numerous To Count

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

LCMS

Prep Batch: 535560

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-27106-1	Field Blank	Total/NA	Water	3535	
190-27106-2	Outfall	Total/NA	Water	3535	
190-27106-3	Outfall Duplicate	Total/NA	Water	3535	
MB 320-535560/1-A	Method Blank	Total/NA	Water	3535	
LCS 320-535560/2-A	Lab Control Sample	Total/NA	Water	3535	
LCSD 320-535560/3-A	Lab Control Sample Dup	Total/NA	Water	3535	

Analysis Batch: 536189

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-27106-1	Field Blank	Total/NA	Water	537 (modified)	535560
190-27106-2	Outfall	Total/NA	Water	537 (modified)	535560
190-27106-3	Outfall Duplicate	Total/NA	Water	537 (modified)	535560
MB 320-535560/1-A	Method Blank	Total/NA	Water	537 (modified)	535560
LCS 320-535560/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	535560
LCSD 320-535560/3-A	Lab Control Sample Dup	Total/NA	Water	537 (modified)	535560

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Client Sample ID: Field Blank

Date Collected: 10/13/21 14:10

Date Received: 10/14/21 09:46

Lab Sample ID: 190-27106-1

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			535560	10/20/21 05:37	EG	TAL SAC
Total/NA	Analysis	537 (modified)		1	536189	10/22/21 14:04	JRB	TAL SAC

Client Sample ID: Outfall

Date Collected: 10/13/21 14:15

Date Received: 10/14/21 09:46

Lab Sample ID: 190-27106-2

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			535560	10/20/21 05:37	EG	TAL SAC
Total/NA	Analysis	537 (modified)		1	536189	10/22/21 14:35	JRB	TAL SAC

Client Sample ID: Outfall Duplicate

Date Collected: 10/13/21 14:17

Date Received: 10/14/21 09:46

Lab Sample ID: 190-27106-3

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			535560	10/20/21 05:37	EG	TAL SAC
Total/NA	Analysis	537 (modified)		1	536189	10/22/21 14:46	JRB	TAL SAC

Laboratory References:

TAL SAC = Eurofins TestAmerica, Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Analyst References:

Lab: TAL SAC

Batch Type: Prep

EG = Eric Gomez

Batch Type: Analysis

JRB = John Barnett

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac- Semiannual PFAS

Job ID: 190-27106-1

Laboratory: Eurofins TestAmerica, Sacramento

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	Identification Number	Expiration Date
Alaska (UST)	State	17-020	02-20-24
ANAB	Dept. of Defense ELAP	L2468	01-20-24
ANAB	Dept. of Energy	L2468.01	01-20-24
ANAB	ISO/IEC 17025	L2468	01-20-24
Arizona	State	AZ0708	08-11-22
Arkansas DEQ	State	88-0691	06-17-21 *
California	State	2897	01-31-22
Colorado	State	CA0004	08-31-22
Florida	NELAP	E87570	06-30-22
Georgia	State	4040	01-29-22
Hawaii	State	<cert No.>	01-29-22
Illinois	NELAP	200060	03-18-22
Kansas	NELAP	E-10375	10-31-21
Louisiana	NELAP	01944	06-30-22
Maine	State	CA00004	04-14-22
Michigan	State	9947	01-29-22
Nevada	State	CA00044	08-31-22
New Hampshire	NELAP	2997	04-18-22
New Jersey	NELAP	CA005	06-30-22
New York	NELAP	11666	04-01-22
Ohio	State	41252	01-29-22
Oregon	NELAP	4040	01-29-22
Texas	NELAP	T104704399-19-13	05-31-22
US Fish & Wildlife	US Federal Programs	58448	07-31-22
Utah	NELAP	CA000442021-12	03-01-22
Virginia	NELAP	460278	03-14-22
Washington	State	C581	05-05-22
West Virginia (DW)	State	9930C	12-31-21
Wisconsin	State	998204680	08-31-22
Wyoming	State Program	8TMS-L	01-28-19 *

* Accreditation/Certification renewal pending - accreditation/certification considered valid.

Eurofins TestAmerica, Michigan

Chain of Custody Record

Introduction

America



190-27106 Chain of Custody



Environment Testing
TestAmerica

☐ SDS or Known Hazard Information Supplied by Client

☐ Discrepancies

☐ Short Hold

☐ Rush ☐ 24 Hr ☐ 2-Day ☐ 3-Day ☐ 5-Day ☐ Other: _____

Receipt Evaluation Performed by: Initials: TEH Date: 10/19/21 Time: 946

Client ID: City of Cadillac
Work Order #: 190-27106

Cooler / Sample Receipt

After hours receipt: complete gray areas. Place cooler in walk-in, place form in Receiving box. Date: _____ Time: _____

Method of Shipment:

Walk-In Client Eurofins TA Field/Courier

Other Client / 3rd Party Courier: _____

Fed Ex Tracking #: _____

☒ UPS Tracking #: Over Night

Other: _____

Shipping Container Type:

☒ Cooler ☐ Box

☐ None ☐ Other: _____

Packing Materials:

☒ Plastic Bags ☐ Foam

☐ Bubble Wrap ☐ Paper

☐ Packing Peanuts ☐ None

☐ Other: _____

Custody Seals Intact:

☒ Yes ☐ No

☐ NA (not used or required)

Cooling Materials:

☐ Ice (Solid) ☒ Ice (Melted)

☐ Blue Ice ☐ None

☐ Other: _____

Bacteriological Samples	Temp Corrected (°C)	Frozen?		Rec'd Within 2 Hrs?		Sample Flagged?	
		Yes	No	Yes	No	Yes	No

Received on same day sampled? Yes ☒ No

Additional Sheets Required? Yes No

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Acceptable	Cooler ID	Affected Samples
<u>CP313207</u>	<u>4.3</u>	<u>4.3</u>		☒	☒ Y ☐ N		
					☐ Y ☐ N		
					☐ Y ☐ N		

Receipt Questions**	Y	N	NA	"No" answers require additional comment
CoC present and ETA receipt signature, date, and time properly documented?	☒			
Containers and Labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	☒			
Appropriate containers used and adequate volume provided?	☒			Preserved bottles checked for pH?* Yes No
Number of sample containers match CoC?	☒			pH strip lot # _____
Samples received within hold?	☒			
Samples submitted for GRO and Volatiles analysis (8260, 624, 524) received without headspace?			☒	
Was a Trip Blank received with VOA samples?			☒	
Were the samples free of any questionable physical conformities? (i.e.; field duplicates or multiple bottles of the same sample do not significantly vary in appearance – color, solid proportions, etc.)	☒			
Were the CoC bottle labels and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	☒			
**May not be applicable if samples are not for compliance testing				*Excludes FOG, VOAs, TOC Vials, HEM

Client Contact Record

Contact Via: ☐ Phone ☐ Email ☐ Other: _____ Person Contacted: _____ Date/Time: _____

☐ Discrepancy allowance agreement is on record in the client project file

Discussion / Resolution

* CoC has 24 analytes, Project list 28, no option for 24. Logged 28 analytes

Any additional documentation and clarification from the client must be noted in the narrative and/or scanned into the CoC directory.

Reviewed by Je Hec Date: 10/14/21

WI-MI-010_020720

